AIAA JOURNAL
Vol. 61, No. 11, November 2023

Study of Side-Jet Interactions over a Hypersonic Cone Flow Using Kinetic Methods

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https://doi.org/10.2514/1.J063020

The interaction between the hypersonic flow over a cone and a side jet creates complex multiple recirculation regions as well as separation, reattachment, and bow shocks known as Edney type-IV shock/shock interactions. The interactions between these multiscale structures are shown to be well captured with the direct simulation Monte Carlo (DSMC) method, which is a particle-kinetic approach. This computational technique was used because even though the freestream conditions were continuumlike, sufficiently strong gradients in the shock and jet expansion were observed to cause continuum breakdown. The time-accurate DSMC approach has been shown in previous work to capture the multiscale structures and unsteadiness in the shock structure that occurs in such flows, as was also observed in this work. These sources of unsteadiness in the shock system are shown to locally affect surface parameters, such as heat transfer and skin friction, as well as predict a low–frequency unsteadiness consistent with earlier values in the literature.

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Nomenclature

A_b	=	base area of the cone, m^2
c_f	=	skin-friction coefficient
c_h	=	heat transfer coefficient
D_b	=	base diameter of the cone, m
F_D	=	drag force acting on cone, N
$FNUM$	=	number of real particles each simulation particle represents
F_N	=	normal component of the resultant pressure force acting on the cone, N
f	=	frequency, Hz
h	=	penetration height of the jet gas, m
j	=	properties of the jet gas
L	=	length of the cone, m
M	=	Mach number
$\widetilde{m}$	=	normalized mass flow rate of the jet gas
$\dot{m}$	=	mass flow rate of the jet gas, kg/s
P	=	continuum breakdown parameter
p	=	pressure, Pa
Re	=	Reynolds number
R_j	=	gas constant of the jet gas, $J/kg \cdot K$
r_b	=	radius of the base of the cone, m
St	=	Strouhal number
T_{0j}	=	stagnation temperature of the jet gas, K
U_∞	=	freestream velocity, m/s
γ	=	ratio of the specific heats
Δt	=	time step or time difference, s
θ_{BS}	=	angle between the bow shock and the cone surface, deg
θ_{SL}	=	angle between the shear layer and the cone surface, deg
λ	=	mean free path, m
ρ	=	mass density, kg/m^3
τ_{flow}	=	flow time, s

Presented as Paper 2022-3987 at the AIAA Aviation 2022 Forum, Chicago, IL and Virtual, June 27–July 1, 2022 and as Paper 2023-0857 at the AIAA SciTech 2023 Forum, National Harbor, MD, January 23–27, 2023; received 22 March 2023; revision received 2 June 2023; accepted for publication 7 July 2023; published online 25 July 2023. Copyright © 2023 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 1533-385X to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

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I. Introduction

BECAUSE side jets are one of the widely used control mechanisms for maneuvering and reentry for high-Mach-number vehicles [1], the flow structure caused by the jet/shock wave interactions and the possible effects of these flow structures on surface properties (such as drag forces and heat transfer) have been the focus of many studies [2–4]. Understanding these phenomena, which are known as aerodynamic interference effects, will aid in the design of the control mechanisms of such vehicles. The specific flow structure and the effect of the jet depends on the flight configuration and the geometry of the vehicle and control surfaces, as well as the flight regime. The effect of a side jet will cause the flow structure to be complex due to separation and separation-related shock waves upstream of the jet, as well as circulation and reattachment of the flow downstream of the jet. The overall characteristics of these flowfields depend on whether the freestream flow is supersonic or hypersonic, whether the incoming boundary layer is laminar or turbulent, and several other parameters such as the penetration height of the jet, the momentum ratio of the jet to the freestream, etc.

A review of the literature shows that most of the previous studies of side jets, or traverse jets, are for lower altitudes of unit Reynolds number Re of $\sim 10^7$ m^{-1}. Figure 1 shows the unit Reynolds number versus the freestream Mach numbers for some of these studies. Most of the earlier work obtained measurements on several different configurations of the jet and test article [2,5–18]; however, a number of numerical studies using continuum methods [19–22] were also performed. The studies shown in Fig. 1 included both laminar and turbulent boundary layers, different pressure ratios of the jet to the freestream, and several jet outlet conditions. In general, it was observed that the Mach number of the incoming boundary layer and the pressure ratio of the jet to the freestream are the two important parameters that affect surface parameters such as drag force and heat transfer. There are also a few studies that use kinetic methods, but almost all of them are for higher altitudes [23–26]. In these earlier works, the main focus was the expansion of a jet to a rarefied environment; in some cases, chemical reactions inside the jet expansion were also considered. However, with the advances in computational algorithms and hardware, we can use the direct simulation Monte Carlo (DSMC) [27] for continuumlike freestream number densities, as indicated by the red diamonds in Fig. 1.

The most important aspect of modeling flows with side jets is the jet expansion because it affects all the flowfield characteristics in the interaction region [26,28]. The jet flow causes a blockage to the incoming flow, which then results in separation of the boundary layer upstream of the jet. The extents of this separation and the height of the jet gas penetration depend strongly on the jet expansion characteristics. For example, the outlet speed of the jet was shown in Ref. [1] to

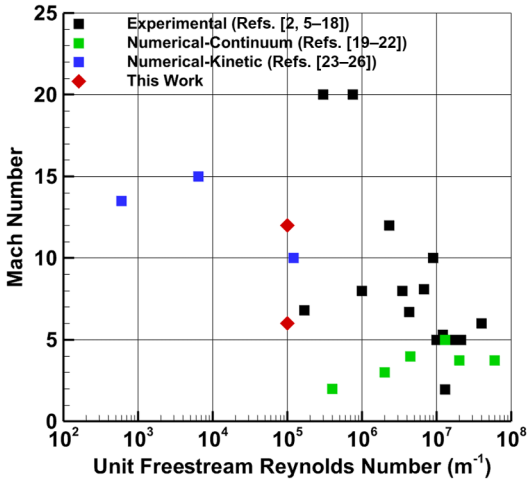

Fig. 1 Freestream Mach and unit Reynolds numbers of the previous related studies.

change the expansion shape and structures of the interaction zone. Moreover, the expansion of the jet occurs in a very small spatial region as compared to the simulation domain, suggesting that there will be high gradients occurring in this region. Because the incoming flow is still supersonic even after the leading-edge shock in hypersonic flows for most cases, the presence of the jet causes several shocks and strong shear layers, regardless of the freestream Reynolds number. The presence of these high-gradient regions will be shown in this work to cause continuum breakdown. Therefore, there is a strong need to use kinetic approaches, such as the DSMC method, to simulate such flows.

DSMC is a well-established time-accurate method to model flows with high gradients because it can capture shocks and shear layers without problems due to numerical viscosity. Previous work with DSMC simulations [29–31] included several geometries such as double-wedge and double-cone geometries as well as compression ramps, which all create separation bubbles and several shocks and shear layers. We have characterized the unsteadiness in the shock system of a double-cone configuration that is caused by shock/boundary-layer and shock/shock interactions [29,30]. We have also discovered instabilities occurring in the shock layer that are in sync with instabilities in the shear layer for the double-wedge flow in Ref. [31]. As is well known, these types of shock/shock and shock/boundary-layer interactions can cause unsteadiness in the entire shock system [32]; however, the specific importance of the shock system versus the boundary layer was previously unknown. The oblique shock/side-jet interaction creates a rich set of shock/shock interactions in a similar manner. Yet to date, the main effort in the literature in modeling side-jet interactions has been to characterize steady-state surface parameters, and only a small number of studies studied unsteadiness of the flowfield and the effect of the unsteadiness on surface parameters [5,6,10,14]. Similar to the previous modeling of laminar shock-shock and shock-boundary layer interactions by Tumuklu et al. [29,30] and Sawant et al. [31], we analyze the unsteadiness of gas-surface parameters in side-jet interactions using the DSMC, an inherently time-accurate method.

In earlier previous modeling of side-jet interactions in a hypersonic flow, Karpuzcu et al. compared the effect of different fundamental geometries [33] (e.g., flat plate, wedge, and cone configurations) for a freestream Mach number of six. We found from these studies that the interaction region was the largest for the flat-plate case, but the surface parameters changed the most for the cone geometry as compared to the case with no side jet. Preliminary experiments have been conducted at Purdue University's Boeing/Air Force Office of Scientific Research (AFOSR) Mach-6 Quiet Tunnel (BAM6QT) quiet tunnel [33,34] with a 7 deg half-angle cone at zero angle of attack. Qualitative agreement between our simulations and the experiments in terms of shock structures and flow unsteadiness was observed, but more detailed comparisons are still required. In this work, we present flowfield and surface forces for the same cone geometry used at the BAM6QT in both the preliminary side-jet experiments [33,34] and transition studies [35–37]. In addition, we consider two different Mach numbers, one that is the same as the experiments at M=6 and the second at M=12, to understand how the flow physics in the interaction region changes. Our primary goals in this work are to investigate the overall flowfield characteristics, surface parameters, unsteadiness in the flowfield due to shock/shock and shock/boundary-layer interactions, and effect of this unsteadiness on the local surfaced parameters.

The organization of the paper is as follows. In Sec. II, we discuss the specific cases in terms of their freestream and numerical DSMC parameters, and we present time-averaged flowfields. Following that, in Sec. III, we present the effect of the jet presence on the steady-state surface parameters. In Sec. IV, we consider the instantaneous flowfield results and characterize the frequencies of the unsteadiness and its effects on surface parameters. Finally, in Sec. V, we summarize our work.

II. Time-Averaged Steady-State Flowfield Results

In this work, hypersonic flows over a cone with a 7 deg half-angle were simulated using the statistical modeling in low-density environment (SMILE) [38] as the computational solver. A sonic jet was placed at the middle of the cone surface for both the Mach 6 and 12 flows with a working fluid of molecular nitrogen for both the freestream and jet gas. The cone has a base diameter and a length of 0.1 and 0.4 m, respectively; and the jet has a diameter of 1 mm. In normalizing the spatial dimensions, we used the cone length, and we put the origin of the flowfield at the cone tip. Thus, the cone started at $x/L = 0.0$ and ended at $x/L = 1.0$, and the center of the jet was located at $x/L = 0.5$ and $y/L = 0.0625$. As mentioned earlier in this paper, the dimensions of the cone were selected to be the same as preliminary experiments conducted at the BAM6QT at M=6 [34]. The freestream and jet flow properties are given in Table 1 for a unit freestream Reynolds number of 1×10^5 m^{-1}.

In terms of numerical parameters, the total number of simulation particles was on the order of 500×10^6, with at least 10 particles in each collision cell locally for both cases with a time step of $\Delta t = 5.0 \times 10^{-9}$ s. The flow was simulated as an *axisymmetric* case with the x axis as the symmetry axis, and the surface of the cone was treated as an isothermal wall at $T_w = 300$ K. The computational domain had a length of 0.425 m in the x direction and a height of 0.15 m in the y direction. The time required for the flow to establish in the interaction region is 2.5 ms, which corresponds to 500,000 time steps. This time is determined by monitoring the numerical probes placed in the flowfield from the beginning of the simulation at time $t = 0$. We start time sampling after this time instant and average for 100,000 time samples, which correspond to 0.5 ms to have adequate time-averaged results. The computations were done on Texas Advanced Computing Center (TACC's) Stampede2 supercomputer. On this supercomputer, a run of 5 ms in physical time required 460,000 CPU hours on Intel Xeon Phi 7250 ("Knights Landing") processors.

SMILE employs the majorant frequency scheme [39] for modeling collisions. To accurately model viscosity–temperature dependence, the variable hard sphere [27] model was used with a nitrogen

Table 1 Freestream and jet flow parameters for the M=6 and M=12 case simulations

	M6		M12	
Parameter	Freestream	Jet	Freestream	Jet
Number density, m^{-3}	1.00×10^{22}	1.65×10^{23}	0.50×10^{22}	8.25×10^{23}
Mach number	6.0	1.0	12.0	1.0
Temperature, K	50.0	250.0	50.0	250.0
Pressure, Pa	6.9	571.2	3.45	2847.4
Knudsen number[a]	0.07	0.0067	0.15	0.0013

[a]Based on the reference length of the jet diameter of $d = 1$ mm.

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molecular diameter of 4.467 Å and a viscosity index of 0.26 at a reference temperature of 273 K. In addition, it was assumed that there were no chemical reactions, which is reasonable, because the total temperature of the freestream and jet flows was lower than 1500 K.

First, we consider the flow features that occur in time-averaged flowfields, as shown in Fig. 2. It can be clearly seen from both Mach number contours with streamlines (Figs. 2a and 2b) and numerical schlieren images (Figs. 2c and 2d) that the characteristic structures in the side-jet interaction zones discussed by Cassel [1] are well captured. Starting with the Mach number contours, two main recirculation zones are identified: one upstream (UR) and one downstream (DR) of the jet. The upstream recirculation zone is caused by the separation of the incoming boundary layer upstream of the jet and consists of two smaller recirculation zones (UR-1 and UR-2). It is also notable that these smaller recirculation regions in the upstream separation zone are unequal in size, which is in agreement with the earlier works of Refs. [1,2]. The main features of the characteristic shock structure surrounding the interaction region is also seen, starting from the leading-edge shock (LS) that is created by the tip of the cone. Due to the separation caused by the presence of the jet, a separation shock (SS) and a separated shear layer (SL) form upstream of the jet. A bow shock (BS) caused by the jet expansion and reattachment shock (RS), where the flow reattaches downstream of the jet, are also seen. These shocks create Edney type-IV shock/shock interactions [32], which are well known to be unsteady, as was shown in earlier DSMC simulations [29–31].

As mentioned earlier, an important consideration of flows dominated by shocks, expansions, and shear layers is the phenomenon of continuum breakdown, or regions in the flow where the continuum assumption is not valid, regardless of the freestream Reynolds number. Bird defined a continuum breakdown (CB) parameter P that depends mainly on the density gradient, the Mach Ma number, and local mean free path; and it is given by [27]

$$P = \sqrt{\frac{\pi\gamma}{8}}\frac{M\lambda}{\rho}\left|\frac{d\rho}{d\bar{x}}\right| \tag{1}$$

According to this CB parameter P, Bird estimated that continuum breakdown occurs when $P > 0.02$. Because the density gradient is an important term when considering continuum breakdown, not only do the rarefied flows have continuum breakdown, but it is seen in any flow with a sufficiently high-density gradient. In Fig. 3, which shows the spatial distribution of the continuum breakdown parameter contours in the interaction zone, it is clear that in the leading-edge, separation, and bow and reattachment shocks, continuum breakdown occurs. In addition, and of even more importance, continuum breakdown occurs in the jet expansion. The jet expands from the sonic outlet to a Mach number of five to six in a very small distance, thus causing very high gradients, even though the jet is at a higher number density than the incoming flow.

We now turn our attention to the separation and reattachment lengths and their relation to the jet penetration height. The definition of the penetration height is given in Fig. 4 following Ref. [40]. The penetration height is defined as the normal distance from the surface to the location where there is a black "streak" between the bow shock and the normal shock of the jet expansion in the schlieren images [10,40]. This streak is an increase in the schlieren intensity where the jet flow is turned by the freestream and is assumed to be the furthest that the jet gas can penetrate into the flow. The results for the M=6 and M=12 cases are presented in Table 2. It can be seen that the primary

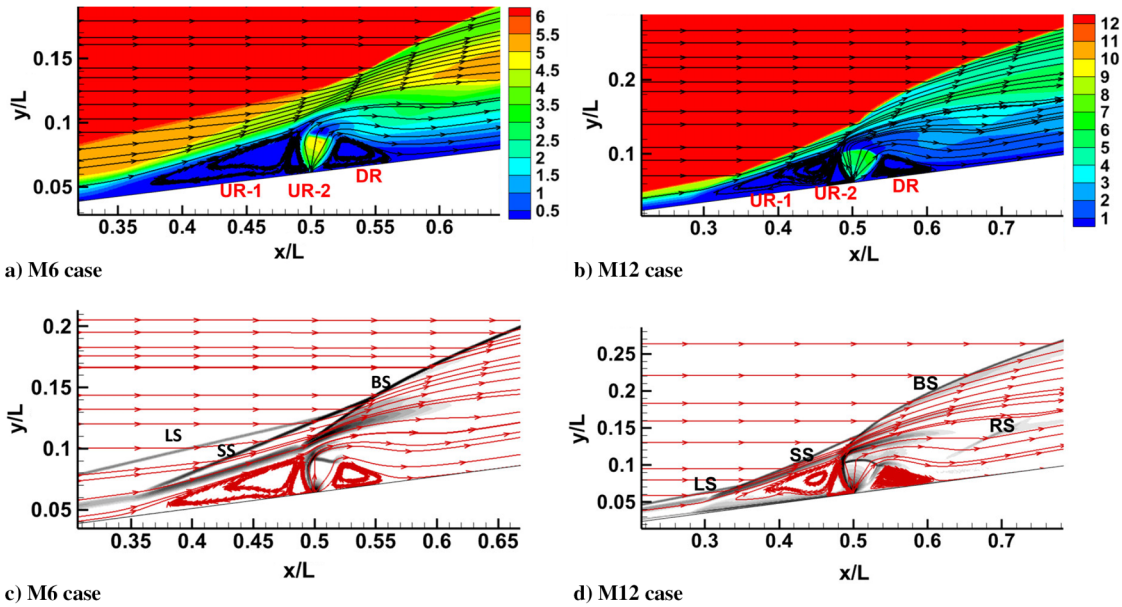

Fig. 2 Contours of Mach number and numerical schlieren for M6 and M12 cases.

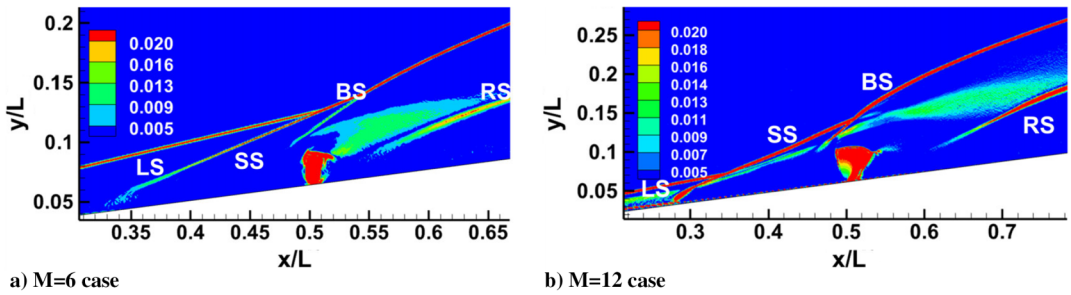

Fig. 3 Contours of continuum breakdown parameter for M6 and M12 cases. The maximum contour level is 0.02, which indicates that continuum breakdown occurs in the red-colored regions.

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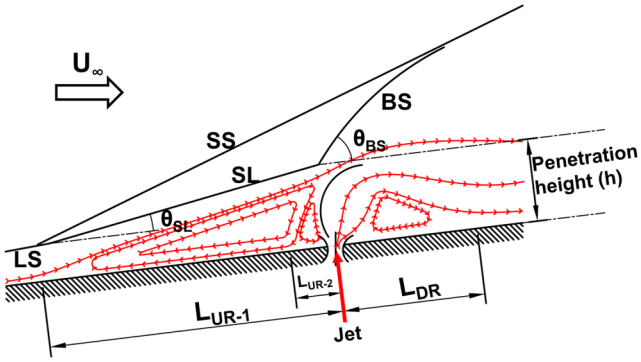

Fig. 4 Sketch of the interaction region showing shear-layer angle θ_{SL}, bow shock angle θ_{BS}, penetration height, and upstream (L_{UR-1}, L_{UR-2}) and downstream L_{DR} separation lengths.

Table 2 Penetration height as well as separation and reattachment distances for the M=6 and M=12 cases[a]

Parameter	M6	M12
Penetration height h, m	0.015	0.025
Primary upstream separation length (L_{UR-1}/h)	4.02	3.32
Secondary upstream separation length (L_{UR-2}/h)	0.67	0.55
Downstream reattachment length (L_{DR}/h)	1.67	1.81

[a]Separation and reattachment distances are nondimensionalized by the penetration height for each case.

upstream separation length L_{UR-1}, defined as the distance along the cone surface from the jet location to the separation point as shown in Fig. 4, is 3.5 to four times the penetration height for both cases. Furthermore, both the secondary separation L_{UR-2} and downstream reattachment L_{DR} lengths scale with the penetration height, as expected, and are less sensitive to the freestream parameters.

Several other flow features such as, dead air zone and bow shock angles [5], penetration height [10] and amplification factor [2], describe the flowfield in the interaction region and allow us to relate our simulations with the existing literature. The definition of these angles and their relationship to the penetration height are again given in Fig. 4. The amplification factor K_F is defined as the ratio of the total interaction force (i.e., thrust from the jet plus surface forces due to the interaction region) to the vacuum jet thrust and is given as

$$K_F = \frac{F_{\text{Interaction}} + F_{\text{Thrust}}}{F_{\text{VacuumThrust}}} \tag{2}$$

First, we compare the nondimensionalized penetration heights that were shown in Table 2 with theoretical relationships taken from Refs. [10,40]. The theoretical penetration height may be expressed as

$$h/r_b = 2\left[\frac{\gamma_j \widetilde{m}}{U_\infty}\right]^{1/2}\left[\frac{2R_j T_{0j}}{\gamma_j(\gamma_j - 1)}\right]^{1/4} \tag{3}$$

where the scaled mass flow rate $\widetilde{m}$ is given by normalizing the jet mass flow rate by the freestream velocity, the density, and the base area of the cone, i.e.,

$$\widetilde{m} = (\dot{m}/4\rho_\infty U_\infty A_b) \tag{4}$$

Then, using the scaled mass flow rate, the theoretical penetration height given by Eq. (3) may be obtained, where the equation is derived from the momentum balance of the jet expansion. Note that to calculate the mass flow rate from the jet, we have used its circumferential outlet area because it is an axisymmetric case. It can be seen from the first row of Table 3 that our results agree very well with the theoretical calculations for the penetration height. Next, we consider angles of the shear layer and bow shock with the cone surface because

Table 3 Comparison of the parameters from DSMC with the literature

	M6		M12	
Parameter	DSMC	Literature [2,5,10]	DSMC	Literature [2,5,10]
Penetration height (h/r_b) [10]	0.32	0.331	0.5	0.523
Shear layer θ_{SL} and bow shock angles θ_{BS} [5]	17 and 33	17 and 36	13 and 31	13 and 33
Amplification factor [2]	5.9	2.5–5.2	2.89	2.5–5.2

they significantly influence shock/shear-layer interactions. Using figure 10 from Ref. [5], we compare the bow shock angles. Note that for each shear-layer angle, a bow shock angle can be calculated from figure 10 of Ref. [5], or vice versa. We take the shear-layer angles from our simulations to calculate the expected bow shock angles. Our results again agree very well with the expected shock angles, as shown in Table 3. Lastly, we compare the amplification factor with those given in Ref. [2]. In terms of amplification factors, most experimental results found in the literature range from supersonic to hypersonic and from laminar to turbulent flows. The most relevant cases for this work are the hypersonic and laminar cases with amplification factors between 2.5 and 5.2. In our two cases, we obtain amplification factors that are higher than this range, mainly due to our use of a jet mass flow rate smaller than the cases in the literature. However, even for the smaller mass flow rate, the jet causes enough blockage to create a large interaction region because our cases are axisymmetric. This has the effect of causing the interaction force to be much higher than that from the jet thrust.

The mixing characteristics of the flow are another important effect created by the presence of the side jet. Figures 5a and 5b show the mole fractions of jet gas for the two Mach number cases where it can be seen that the only mixing zones are the upstream separation zones, UR-1 and UR-2, in both cases. The mole fraction contours demonstrate that there is no mixing downstream of the jet and that the boundary layer only consists of the jet gas. To further illustrate the lack of mixing between the jet and freestream gas, translational temperature contours are shown in Figs. 5c and 5d, where it can be seen that the jet expands to very low temperatures for both flow Mach number cases. The temperatures inside the upstream separation region are much lower than the freestream total temperatures ($T_0 = 420$ K for the M6 case, and $T_0 = 1500$ K for the M12 case) because of the mixing of the cold jet gas with the incoming hot freestream gas. The translational temperature of the jet gas is lower than the isothermal wall temperature of 300 K in the boundary layer downstream of the jet because there is no mixing and only the jet gas exists in the boundary layer. In Figs. 5e and 5f, it can be seen that temperatures decrease toward the cone surface for the upstream locations but increase toward the wall for the downstream locations for both the M6 and M12 cases. This means that to satisfy the isothermal wall boundary condition downstream of the jet, the flow must remove energy from the wall. This will have implications for the stability of the boundary layer downstream of the jet because there is energy addition to the flow that might eventually cause transition to turbulence, but that is beyond the scope of this work.

III. Steady-State Gas–Surface Interaction Results

An important objective of a side jet is to create forces on control surfaces. However, in a hypersonic incoming flow, complex flow interactions due to the presence of the jet can create surface forces and influence the local heat transfer values in addition to the force from the jet itself. Therefore, for design purposes, surface parameters such as pressure distributions, heat fluxes, and friction forces in addition to the thrust force created by the jet are useful.

We first compare our simulations at M=6 with surface pressure distributions from the experiments of Spaid and Zukoski [41] and

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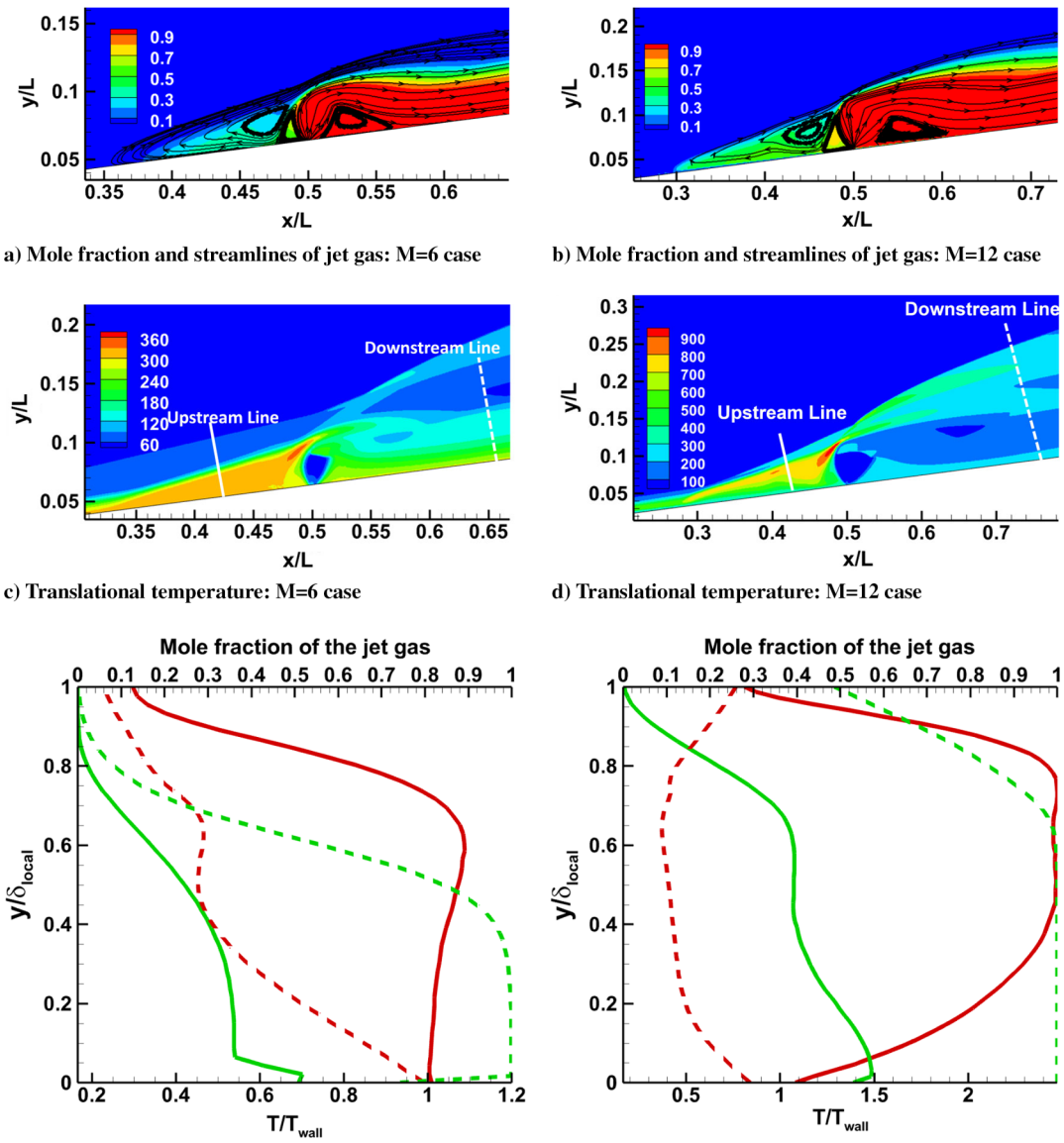


a) Mole fraction and streamlines of jet gas: M=6 case

b) Mole fraction and streamlines of jet gas: M=12 case

c) Translational temperature: M=6 case

d) Translational temperature: M=12 case

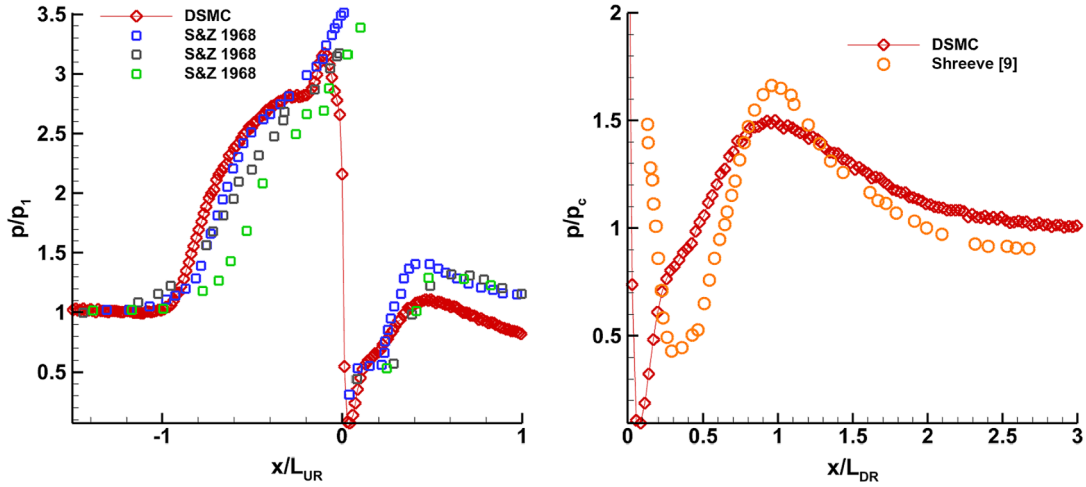
e) Translational temperature and mole fraction profiles: M=6 case

f) Translational temperature and mole fraction profiles: M=12 case

Fig. 5 Mole fraction of jet gas and translational temperature contours. Last row shows translational temperature and mole fraction profiles along white lines shown in middle-row figures at locations upstream and downstream of jet. Solid lines are upstream location, dashed lines are downstream location, red lines are temperatures, and green lines are mole fractions.

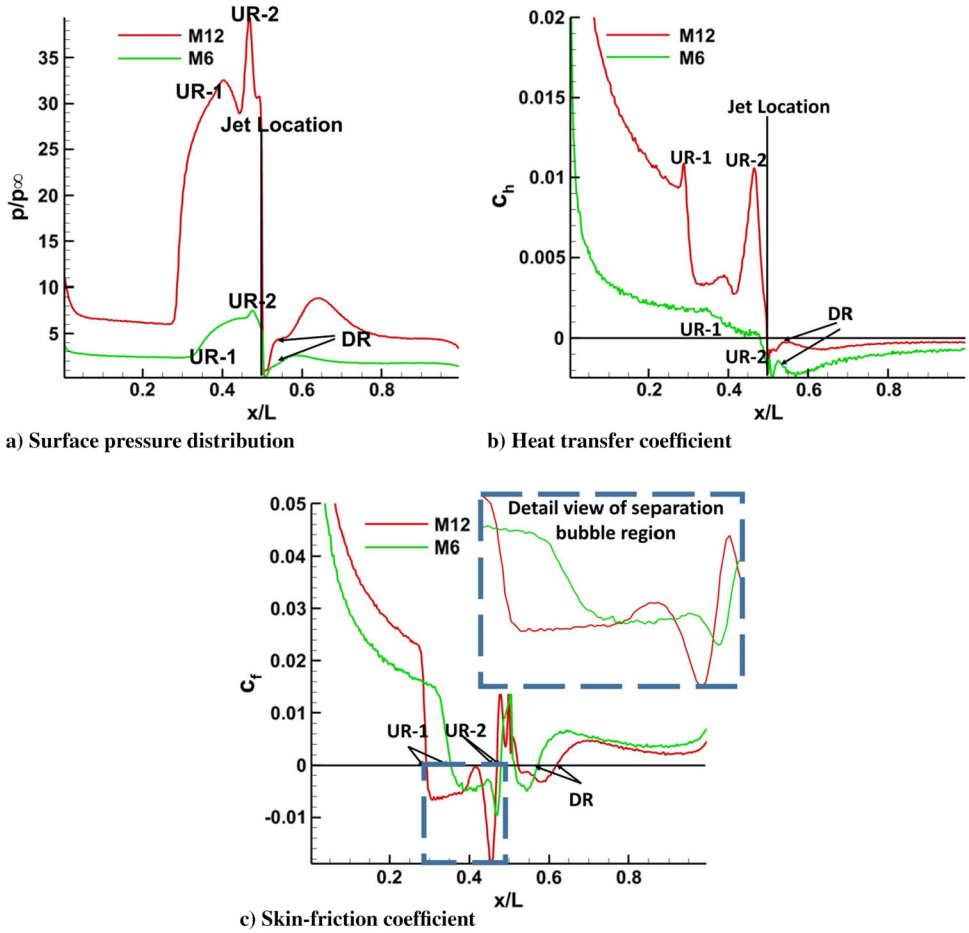
Shreeve et al. [9] for two-dimensional (2-D) and axisymmetric configurations, respectively, in Fig. 6. Note that in Fig. 6, the jet location is offset to the origin of the x axis, i.e., $x = 0$. In these experiments, the jet is either a slit along the transverse direction or a slit along the surface of the cone, thus creating nominally 2-D or axisymmetric interactions. Because our simulations are also for an axisymmetric setting, these provide the best comparison in terms of the flow topology. The incoming Mach number immediately upstream of the jet in the work of Spaid and Zukoski [41] is 4.54, which is comparable to our case of 5.1. Three sets of pressure distributions shown in Fig. 1 for different jet strengths that are between 60 and 200 are compared with our case, which has a value of 100. The jet strength is defined in both of these experiments as the ratio of the jet total pressure to the static pressure of the incoming flow just upstream of separation. In Fig. 6a, we normalize the x axis by the upstream separation length L_{UR}, which is determined as the first point that the surface pressure starts to rise in the experimental data. Surface pressures from DSMC are normalized by the static pressure just upstream of the separation for each case p_1, as was done in the paper. The experiments of Shreeve et al. [9] are for a 5 deg half-angle cone at M=6 freestream flow (the same Mach number as our case) and for a jet strength of 42. Only pressure data from downstream of the jet are available for this case. In Fig. 6b, we normalize the x axis by the downstream separation length L_{DR}, which is determined as the point at which the surface pressure peaks in the downstream of the jet. Surface pressures from DSMC are normalized by the static pressure far downstream of the jet p_c, as was done in the paper. It is clear from Fig. 6 that the pressure distributions obtained from the DSMC simulations agree quite well with experimental results. In addition to the distributions, the plateau and peak pressures for the upstream as well as the smallest pressure downstream of the jet agree well. These results are as expected with appropriate scaling of the parameters and similar flow topology, even though the experiments were conducted at much higher Reynolds numbers.

Having shown reasonable agreement with the available experimental data in the literature, we now show additional surface parameters for both M=6 and M=12 cases. It should be noted that in Fig. 7, the x axis scaling is again with the length of the cone; and the origin of the coordinate system is at the tip of the cone so that the jet is at $x/L = 0.5$, as it was in all the previous figures except Fig. 6. It can be clearly seen from the surface parameters in Fig. 7 that due to the stronger flow interaction region for the M12 case, the maxima of all surface



a) Comparison with Spaid and Zukoski [41] b) Comparison with Shreeve et al. [9]

Fig. 6 Comparison of surface pressures for M=6 case with the experimental results available in the literature [9,41]: DSMC simulations of this work (diamonds), and Shreeve et al. [9] (circles). Also, Spaid and Zukoski [41] is denoted as "S&Z 1968" in the legend.



a) Surface pressure distribution b) Heat transfer coefficient

c) Skin-friction coefficient

Fig. 7 Time-averaged pressure, heat transfer, and skin-friction distributions for M=6 and M=12 cases.

parameters are much higher than the M6 case. The surface pressure plateaus at $x/L = 0.27$ and $x/L = 0.61$ are due to the upstream and downstream separation regions, which are at much higher pressure values for the M12 case as compared to M6. Also, in Fig. 7b, there are two heating peaks in the upstream separation region for the M12 case that are not present in the M6 case. This is mainly due to the fact that the total temperature for the M12 case is higher and the interaction of the separation shock with the leading-edge shock and bow shock creates higher temperatures than the wall temperature. More interestingly, the heat transfer rate downstream of the jet location is negative in both cases, meaning that the heat transfer is from the surface to the flow. This is due to the expansion of the jet and its mixing characteristics, which were shown in Fig. 5, where the boundary layer downstream of the jet only consists of the "cold" jet gas. Lastly, the skin-friction coefficient values are the most similar parameter for the two cases, but as expected, the locations where the coefficient changes sign are different. The shear stress distributions inside the primary upstream separation zone (UR-1) are consistent with those observed previously in the large separation zones in compression corners for supersonic to hypersonic Mach conditions where two minima were found to be present [42–44].

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The integral quantities of the aforementioned surface parameters, total drag force, normal force, and heat transfer were obtained by integrating the local skin friction, pressure distributions, and heat transfer rate over the surface of the cone. Comparisons are made for the integral force quantities for the cases with and without the jet present as well as by excluding the jet thrust force from the jet itself to understand the effect of aerodynamic forces only. Table 4 shows the ratio of the forces with and without the jet for both drag and normal forces. It is clear that the presence of the jet interaction region decreases the overall drag on the cone. Moreover, due to the high-pressure zones created by the separation regions, the normal force increases to almost twice the value of the case with no jet present for both the M6 and M12 cases. More interestingly, we see that the total heat transfer rate decreases; and for the M6 case, it is actually a negative value, as shown in Table 5. This means that the heat transfer is from the wall to the flow; even for the M12 case, the jet causes the overall heat transfer to drop by 80% of the case with no jet present. One advantage of using DSMC is that we can track how much of these surface properties are due to each gas. In Table 5, we show that the heat transfer due to the jet gas is much smaller (for the M6 case, it is even negative) as compared to the freestream gas.

Table 4 The ratio of surface forces with and without a side jet present. The quantities given here are the integral quantities over the cone surface

Force	Formula	M=6	M=12
Drag force ratio	$F_{\mathrm{Djet}}/F_{\mathrm{Dnojet}}$	0.76	0.68
Normal force ratio	$F_{\mathrm{Njet}}/F_{\mathrm{Nnojet}}$	2.30	1.74

Table 5 Heat transfer rates with and without a side jet present[a]

Jet presence	Property	M=6	M=12
Without jet	Total heat transfer rate W	12.6	249.2
With jet	Total heat transfer rate W	−5.68	59.3
With jet	Heat transfer rate due to jet gas W	−8.8	4.9
With jet	Heat transfer rate due to freestream gas W	3.12	54.4

[a]The quantities given here are the integral quantities over the cone surface.

IV. Instantaneous Flowfield Results and Unsteadiness

Up to this point, we have discussed the effect of the jet on time-averaged steady-state flowfields. As mentioned earlier, however, we anticipate that there will be a significant amount of unsteadiness present in the strong shock system, shear layers, and mixing zones. Therefore, we investigate the instantaneous flowfields and time-resolved probe data taken from strategic flow locations.

We first consider pressure fluctuations in the flow by calculating the percent difference between two time intervals. The first time instant (i.e., the base flow at $t_1 = 3$ ms) occurs at a time after the flow establishes, which is also an instantaneous flowfield. The percentage deviation from the base flow at subsequent time intervals is obtained by subtracting the flowfields at $t_1 + \Delta t$, $t_1 + 2\Delta t$, and $t_1 + 3\Delta t$ from the base case t_1, where Δ t is 0.05 and 0.015 ms for the M6 and M12 cases, respectively. Figure 8 shows the percent difference in the pressure values for these three time instants for both cases. It can be clearly seen that deviations are on the order of 20% and higher in the separation, bow, and normal shocks due to the presence of the jet. It is also observed that in the boundary layer downstream of the jet, fluctuations are even higher than the upstream for both cases, with even more structure for the M12 case that resembles a vortex shedding pattern.

Next, we investigate the detailed time variations of pressure fluctuations using the DSMC data taken from numerical probes placed at strategic locations in the flowfield. We place the probes in the locations where shock/shock, shock/boundary-layer, and shock/shear-layer interactions occur, as well as at locations near the separation–reattachment

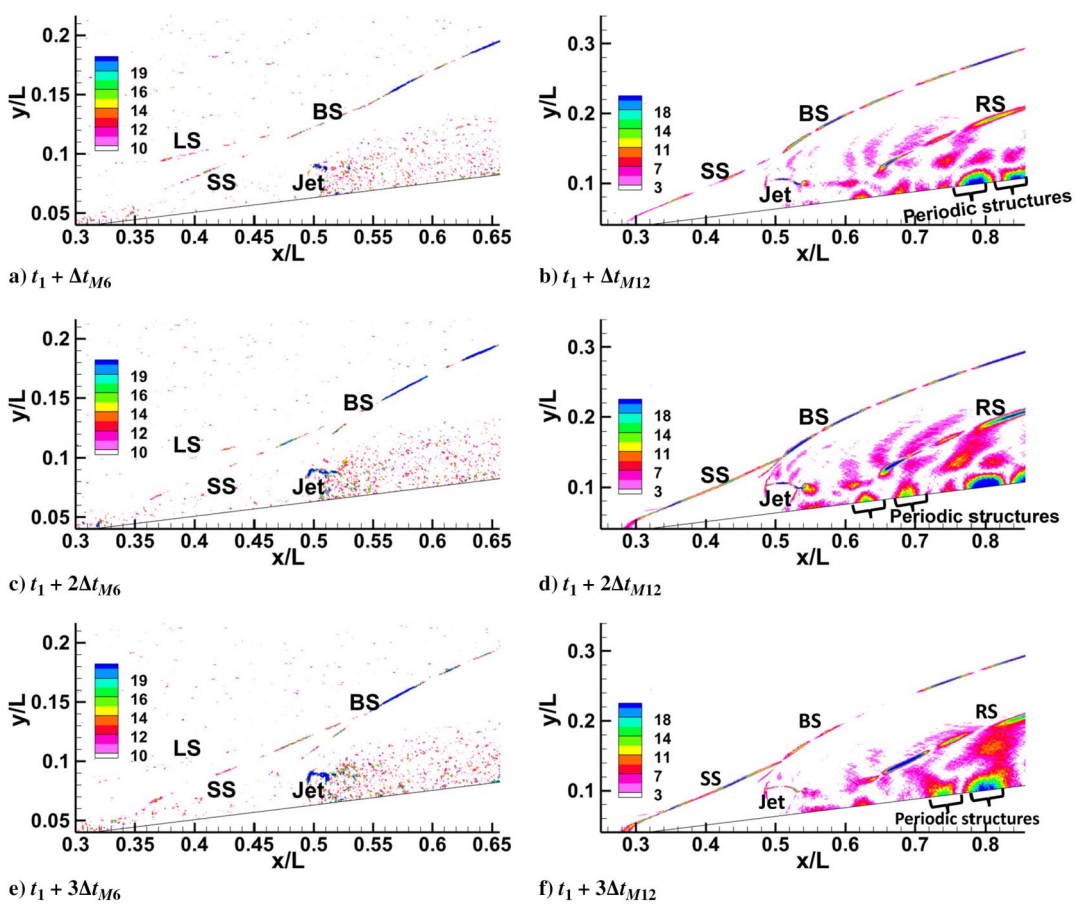
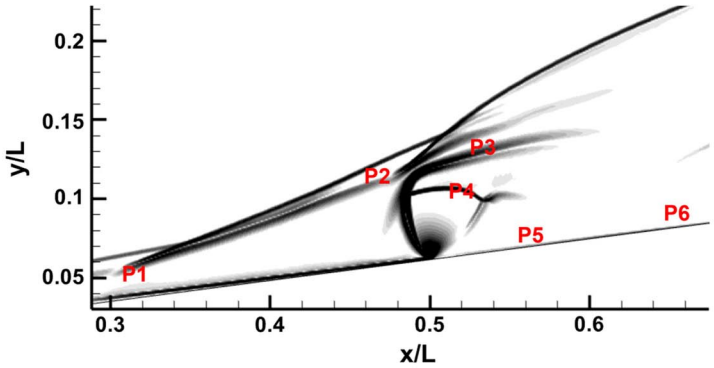
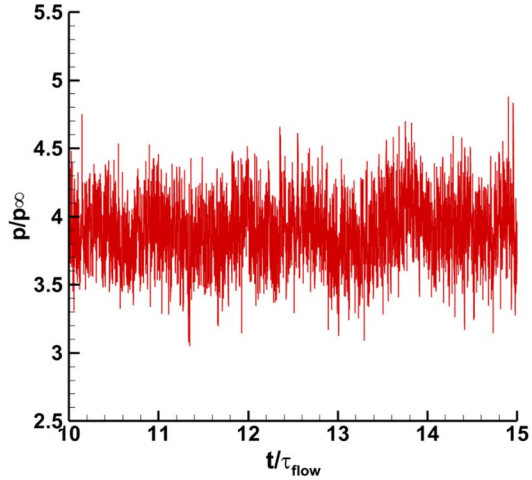


Fig. 8 Percent difference in pressure in instantaneous flowfields with indicated shock structures: M6 case (left column), and M12 case (right column).

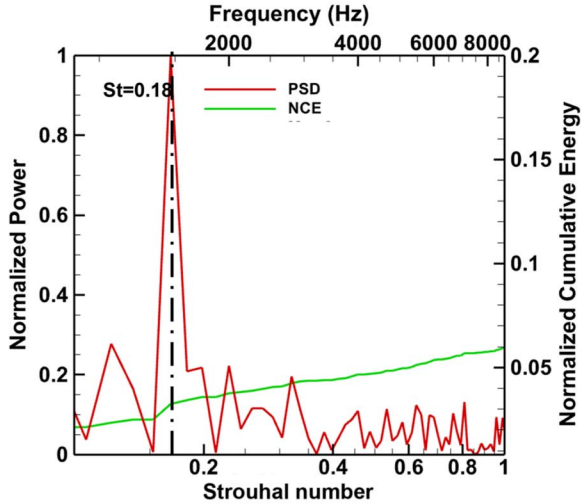
points, as shown in Fig. 9a. In the other subfigures of Fig. 9, we show the time-resolved pressure and its corresponding power spectral density (PSD) plots for the M6 and M12 cases at probe P1 located near the separation shock foot. Similar time behavior was observed for the probes at other interaction regions too, but that is not shown here. Power spectral densities were calculated for the mean subtracted pressure at these locations. Time-resolved DSMC pressure values were collected at a frequency of 200 MHz (i.e., at each time step of 5 ns), and the sampling was performed for 10 ms. Then, using MATLAB'S fast Fourier transform algorithm [45], the spectral content of the mean subtracted pressure data is extracted. Because DSMC is a particle method, there are always periodic fluctuations due to particle collisions. This is due to two sources: one physical (i.e., the thermal speed of the particles), and one numerical (i.e., the DSMC numerical parameter $FNUM$, which is the number of real particles each computational particle represents). The former can never be removed, regardless of



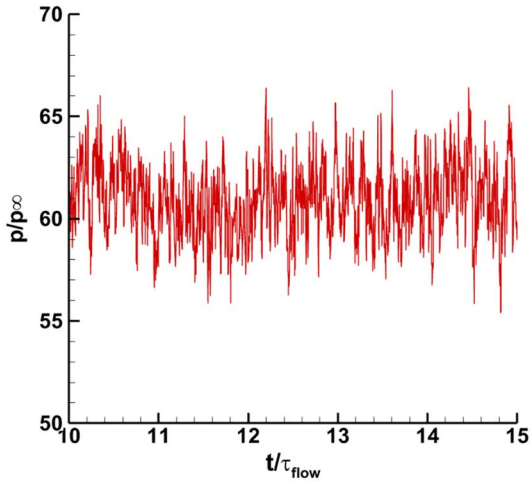
a) Probe locations P1–P6, shown on the schlieren image for M12 case, but the same interaction locations are used for M6 case



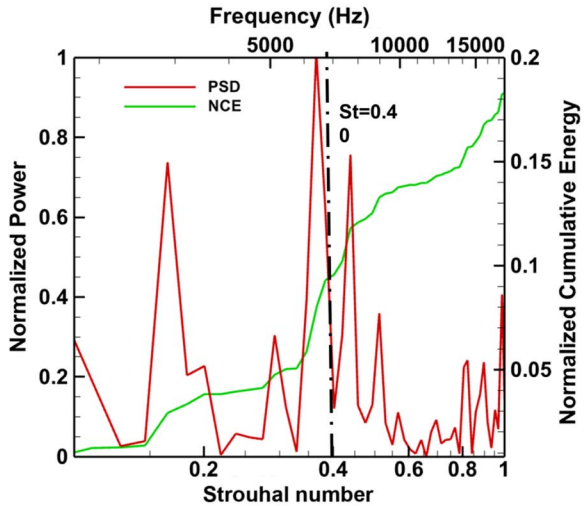
b) Pressure at separation point: M6 case



c) PSD of pressure at separation point: M6 case



d) Pressure at separation point: M12 case



e) PSD of pressure at separation point: M12 case

Fig. 9 Pressure signal and corresponding power spectral densities with normalized cumulative energies taken at the separation shock foot: probe location P1.

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the number of computational particles; whereas the second can be minimized by using as small an FNUM as possible. In Ref. [46], it was shown that the peaks in the PSD due to DSMC noise decrease as $FNUM$ is decreased; i.e., the number of simulation particles are increased but other frequencies due to the physical, thermal nature of the flow persist. It can be seen that fluctuations that have lower frequencies that are lower than the expected DSMC numerical and thermal fluctuations (on the order of megahertz) exist in both cases in the current study too; see Figs. 9c and 9e. The power spectral density analysis of the probe DSMC simulation results show broadband low-frequency bands centered around $St = 0.19$ and $St = 0.4$ for the M6 and M12 cases, respectively, where the Strouhal number and the flow time are defined as

$$St = \frac{fD_b}{U_\infty}, \tau_{\text{flow}} = \frac{L}{U_\infty} \tag{5}$$

These frequencies are near the peak frequencies for each PSD and the inflection point in the normalized cumulative energy (NCE) curves, where the latter is calculated by cumulatively summing up the PSD spectrum. Most importantly, these frequencies are consistent with Strouhal numbers based on cone diameter and freestream velocity, which are found in the literature for cone geometries; see Fig. 10. It appears that the frequency of the unsteadiness in the shock system for the flows with these types of interaction regions increases with Mach number and does not strongly depend on the freestream Reynolds number as long as the incoming boundary layer is laminar. Moreover, there is evidence that the spectral content of the shock oscillations is insensitive to the freestream Reynolds number but depends on the Mach number for canonical cylinder wake flows too [47–49].

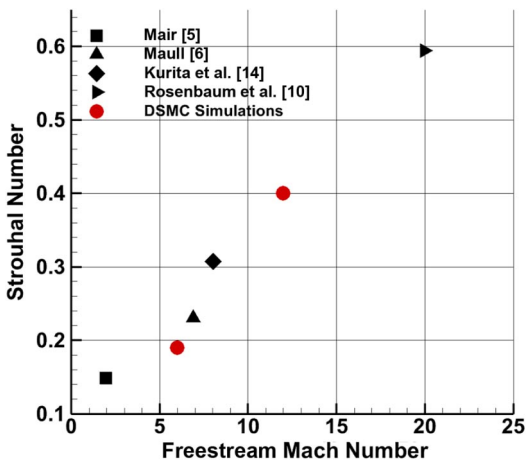
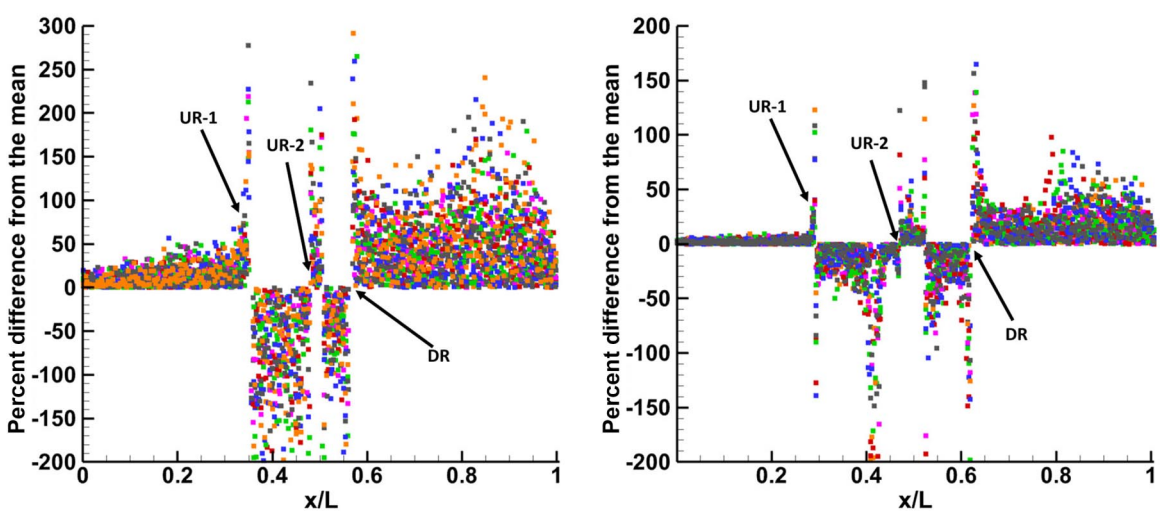


Fig. 10 Comparison of Strouhal number *St* obtained from this work (red circles) with the literature. Note that unit freestream Reynolds numbers *Re* (1/m) are on the order of 10^7 for Mair [5], Maull [6], and Kurita et al. [14]; and they are on the order of 10^5 for Rosenbaum et al. [10] and this work.

The low-frequency unsteadiness in the flowfield also causes unsteadiness of the local instantaneous surface parameters. To study the effect of this unsteadiness on surface properties, the deviation of the local skin-friction coefficient at instantaneous flowfields was calculated at 10 different time intervals for both cases. The first time interval was taken at 10 flow times for both cases; and the last one was at 11.4 and 10.7 flow times, respectively, for the M6 and M12 cases. The difference between these time instants was chosen as 1/10th of the frequencies observed in Figs. 9c and 9e so that at least one cycle of the shock oscillations was resolved. Each color in Fig. 11 shows a data point from one of the 10 time intervals. The skin-friction coefficient plots in Figs. 11a and 11b show very high deviations from the mean, especially near the separation and reattachment points. This does not mean that the actual value of the skin friction is vastly different but rather the separation and reattachment locations, where the skin friction is zero, are not stationary. Moreover, the percent deviation from the mean is 50–100% inside the separation and reattachment regions, which is much higher than the expected DSMC computational noise (5–10% for this sampling rate). Another important feature of the plots in Fig. 11 is that the deviation from the mean is much higher in the downstream boundary layer as compared to the upstream boundary layer; this is consistent with Fig. 5, which showed that the downstream boundary layer consists of purely the jet gas with vortex-shedding-like motion due to the jet expansion, as observed in Figs. 8b, 8d. and 8f. Both of these flow features indicate high unsteadiness.

V. Conclusions

The interactions between a side jet and hypersonic flows at M=6 and M=12 over a slender cone were simulated using the DSMC method. Both the time-averaged steady-state and time-resolved results of shock structures and surface parameters were examined. It was demonstrated that continuum breakdown occurs in the leading-edge shock, the bow shock, and especially in the jet expansion, which motivates the use of a kinetic method. Steady-state results showed that the presence of the jet on the cone increases the complexity of the flowfield by creating a complex shock structure, recirculation regions, and shear layers. It was demonstrated that many of the key normalized parameters, such as penetration height and shock angles, are captured well by the DSMC simulations and agree with theoretical and experimental results. It was shown that the jet and freestream gases mix only in the upstream separation region, and the boundary-layer flow downstream of the jet consists only of the jet gas, resulting



a) Shear stress: M6 case

b) Shear stress: M12 case

Fig. 11 Percent deviation from the mean values of local shear stress. Each color represents a different time instant.

in negative heat transfer values downstream of the jet. Integral values for the surface parameters showed that drag force and heat transfer decrease, and force in the direction normal to the freestream increases with the presence of the jet for both Mach number cases. Instantaneous DSMC flowfields showed high-pressure fluctuations, especially around the shocks and jet expansions with a low-frequency unsteadiness consistent with the literature that can be attributed to these fluctuations. These fluctuations in the shocks and boundary layer also affected the surface parameters: especially the drag force, where deviations on the order of 100% were observed in the jet interaction region.

Acknowledgments

The computational resources for this research are provided by the National Science Foundation's Texas Advanced Computing Center (TACC) Stampede2 supercomputer and Purdue University's Anvil supercomputer, which is a part of Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) (previously Extreme Science and Engineering Discovery Environment (XSEDE)), with project number TG-PHY210065. the authors are grateful to Joseph Jewell from Purdue University for providing the case setup and extensive discussions on the subject.

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